



Aditya

AI Developer | Multi-Agent Orchestration | RL & Cognitive Architectures | IEEE Chair 2026

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Profiles

🔗 **Linkedln**
[Aditya Patil](#)

Summary

Highly technical AI Developer & IEEE Student Branch Chair (2026) with a focus on Multi-Agent Orchestration, Reinforcement Learning, and Applied Cognitive Architectures. Proven track record of building production-ready AI environments (SupportEnv) and long-term memory systems (AISERS). Expert in full-stack development using React, FastAPI, and Supabase, combined with strategic leadership of high-impact technical communities.

Education

PES Pre University College Science (PCMCs)	PUC • Distinction Shivamogga • 2 Years
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<https://pestrust.edu.in/pesitm/>

Jain College of Engineering and Research Computer Science and Engineering (AIML)	B.Tech • 7.7 CGPA Belgaum • 3 years
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<https://jcer.in>

Volunteer

IEEE NKSS North Karnataka Sub-Section	2025-2026
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Worked as **Co-Lead in technical committee**, Students Association Committee (SAC) under IEEE North Karnataka Subsection (**NKSS**), organizing and managing technical activities, student life and volunteer functions throughout the region.

Skills

</> Languages Python, SQL, Bash, C/C++, TypeScript, JavaScript	📄 AI/ML Frameworks LangChain, Transformers, HuggingFace, Scikit-Learn, Reinforcement Learning, CrewAI	🔗 Full-Stack & Backend React 18, Vite, FastAPI, Streamlit, Node.js, Supabase, Flask
🔗 GitHub	📄 LaTeX	

Projects

MemoryForm
AI Long-Term Memory Engine

Developed a multi-agent cognitive architecture using **FastAPI, React, and Supabase** to provide AI agents with persistent, infinite-context memory. Integrated **Sentence-Transformers** and a custom weighted ranking algorithm (Similarity, Recency, Confidence) to solve LLM state-loss.

Tech Stack: React 18, FastAPI, Supabase (Vector Store), Docker, Sentence-Transformers (all-MiniLM-L6-v2), OpenRouter.

Bhasha AI
Multi-Agent Translation Pipeline
Architected a 6-agent translation workflow using **LangGraph** that handles technical glossary injection (Domain Optimization), cultural adaptation, and sentiment alignment for high-fidelity localization.
Tech Stack: LangGraph, LangChain, Transformers.

Disaster Management
Multi-Agent Emergency Orchestrator
Implemented a **CrewAI-based** response system coordinating five specialized agents for incident triage, casualty estimation, and resource management. Used **BART and RoBERTa** for real-time severity classification from live reports.
Tech Stack: CrewAI, Streamlit, Hugging Face (BART, DistilBERT, RoBERTa), Pandas.

SupportEnv
Customer Support RL Environment
Built a production-grade Reinforcement Learning environment to train AI agents on high-stress support tickets, developing a modular sentiment-and-accuracy grader to provide stable reward signals for complex de-escalation workflows.
Tech Stack: FastAPI, RL Core (Stable Baselines3), Gradio UI, OpenEnv Specification.

Visualization
WSN Sensor Deployment Framework
Developed a research-focused optimization tool using Gradient Field Greedy algorithms and Q-Learning to ensure 95%+ sensor coverage while maintaining connectivity thresholds for maximum network longevity in non-uniform terrains.
Tech Stack: Python, Streamlit, Q-Learning (RL), KD-Tree, LEACH/HEED Engines.

Cloud Research
Hybrid Cost-Optimization Engine
Achieved a 28% reduction in cloud over-provisioning costs by designing a hybrid predictive framework that accurately classifies machine workload types and automates scaling for large-scale data centers.
Tech Stack: Python, LSTM, ARIMA (Hybrid Engine), K-Means Clustering, Google Cluster Trace Data.

Elevare
AI-Driven Job Matching Platform
Engineered an end-to-end job matching platform featuring an automated ATS scoring engine and domain-specific recommendation system to provide personalized career roadmaps and real-time job opening updates.
Tech Stack: Vite, React, Flask, Gemini 2.5 Flash API, Supabase (Auth/DB), Vercel.

Paper2Code (ResearchGap Analyzer)
Designed an automated research productivity pipeline that analyzes scientific PDFs to identify methodological gaps and generate functional Python implementations to bridge identified technical deficiencies.
Tech Stack: Streamlit, Gemini Pro API, Patent Extractor Module, Regex Parsing.

SIH (Agri-Sethu)
Smart Crop Recommendation System
Developed a precision agriculture assistant for regional crop and fertilizer advice, integrating geospatial mapping to reduce fertilizer waste and improve yield prediction accuracy by 18%.
Tech Stack: Streamlit, Scikit-learn (Ensemble), MongoDB, SoilGrids API, ISRO Bhuvan Maps.

Arogya Saathi
Rural Health Companion
Built a multilingual healthcare assistant with Offline AI support for rural regions, providing digital medical triage, symptom assessment, and hospital recommendations without requiring high-speed internet.
Tech Stack: Streamlit, LangChain, BioMistral-7B, Ollama (Gemma 3), spaCy (NLP).

Livestack
Livestock Lifecycle Intelligence Engine
Designed an extensible intelligence engine for lifecycle tracking and health monitoring of livestock, utilizing data-driven monitoring blueprints to optimize farm efficiency and productivity.
Tech Stack: Python, Planning Blueprints, AI Analytics.

Experience

JCER IEEE STB CHAIR	Belgaum 2026-2027
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Leading a student branch for 2026 as the **CHAIR** for the student branch.

IEEE Bangalore Section SAC Technical Committee (Member)	2026-2027
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Working as **volunteer in technical committee**, Students Association Committee (SAC) under IEEE Bangalore Section, organizing and managing technical activities, student life and volunteer functions throughout the region.

Languages

English	Kannada	Hindi
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Publications

Beyond Passwords: The Essence and Impact of Multi-Factor Authentication in Cybersecurity IEEE	24-May-2024
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This study explores MFA vulnerabilities, bypass techniques, and modern security solutions using adaptive AI.
<https://ieeexplore.ieee.org/document/11140909>

AR and AI for Driver Assistance and Navigation
A comprehensive study on the synergy between Spatial Computing (AR) and Computer Vision (AI) aimed at revolutionizing Next-Gen Head-Up Displays (HUDs) and Autonomous Driving Systems for enhanced road safety and traffic efficiency.